

**MOTOROLA**

# 1.1 GHz Dual Modulus Prescaler

The MC12028A can be used with CMOS synthesizers requiring positive edges to trigger internal counters such as Motorola's MC145XXX series in a PLL to provide tuning signals up to 1.1 GHz in programmable frequency steps.

The MC12028B can be used with CMOS synthesizers requiring negative edges to trigger internal counters.

A Divide Ratio Control (SW) permits selection of a 32/33 or 64/65 divide ratio as desired.

The Modulus Control (MC) selects the proper divide number after SW has been biased to select the desired divide ratio.

**NOTE: The "B" Version Is Not Recommended for New Designs**

- 1.1 GHz Toggle Frequency
- MC12028A for Positive Edge Triggered Synthesizers
- 6.5 mA Maximum,  $-40$  to  $85^{\circ}\text{C}$ ,  $V_{CC} = 5.5$  Vdc
- Modulus Control Input Level Is Compatible With Standard CMOS and TTL
- Low-Power 4.0 mA Typical

## FUNCTIONAL TABLE

| SW | MC | Divide Ratio |
|----|----|--------------|
| H  | H  | 32           |
| H  | L  | 33           |
| L  | H  | 64           |
| L  | L  | 65           |

**NOTES:** 1. SW: H =  $V_{CC}$ , L = Open. A logic L can also be applied by grounding this pin, but this is not recommended due to increased power consumption.  
2. MC: H = 2.0 V to  $V_{CC}$ , L = GND to 0.8 V.

## DESIGN GUIDE

| Criteria                        | Value | Unit |
|---------------------------------|-------|------|
| Internal Gate Count*            | 67    | ea   |
| Internal Gate Propagation Delay | 200   | ps   |
| Internal Gate Power Dissipation | 0.75  | mW   |
| Speed Power Product             | 0.15  | pJ   |

**NOTE:** \* Equivalent to a two-input NAND gate

## MAXIMUM RATINGS

| Characteristic               | Symbol    | Range           | Unit               |
|------------------------------|-----------|-----------------|--------------------|
| Power Supply Voltage, Pin 2  | $V_{CC}$  | $-0.5$ to $7.0$ | Vdc                |
| Operating Temperature Range  | $T_A$     | $-40$ to $85$   | $^{\circ}\text{C}$ |
| Storage Temperature Range    | $T_{stg}$ | $-65$ to $150$  | $^{\circ}\text{C}$ |
| Modulus Control Input, Pin 6 | MC        | $-0.5$ to $6.5$ | Vdc                |

**NOTE:** ESD data available upon request.

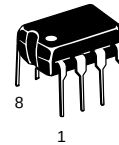
# MC12028A MC12028B

## MECL PLL COMPONENTS $\div 64/65$ , $\div 128/129$ DUAL MODULUS PRESCALER

### SEMICONDUCTOR TECHNICAL DATA

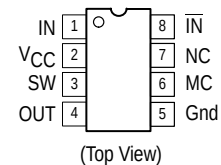


**D SUFFIX**  
PLASTIC PACKAGE  
CASE 751  
(SO-8)



**P SUFFIX**  
PLASTIC PACKAGE  
CASE 626

## PIN CONNECTIONS



## ORDERING INFORMATION

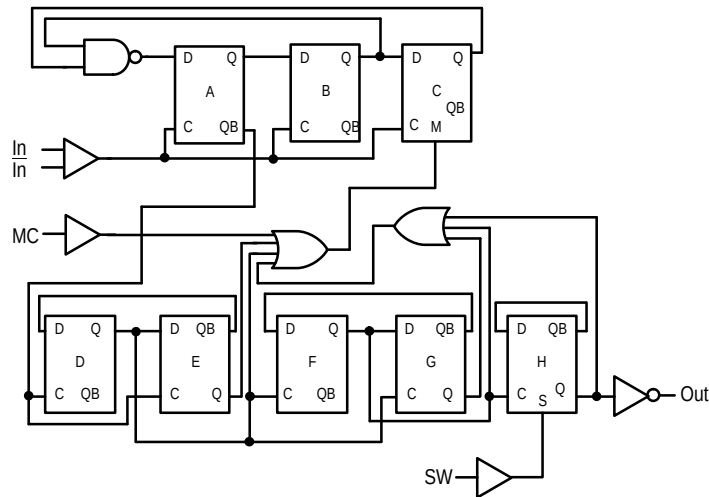
| Device    | Operating Temp Range                         | Package |
|-----------|----------------------------------------------|---------|
| MC12028AD | $T_A = -40^{\circ}$ to $+85^{\circ}\text{C}$ | SO-8    |
| MC12028AP |                                              | Plastic |
| MC12028BD |                                              | SO-8    |
| MC12028BP |                                              | Plastic |

# MC12028A MC12028B

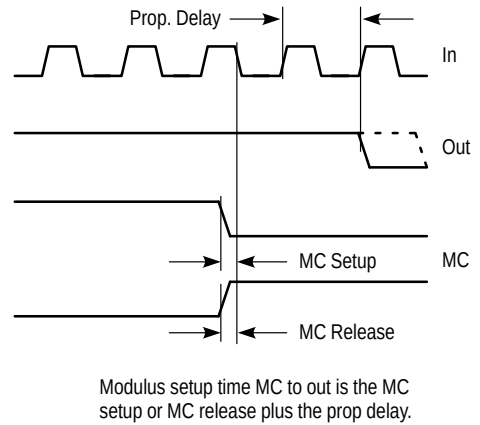
**ELECTRICAL CHARACTERISTICS** ( $V_{CC} = 4.5$  to  $5.5V$ ;  $T_A = -40$  to  $85^\circ C$ , unless otherwise noted.)

| Characteristic                                                 | Symbol    | Min        | Typ      | Max          | Unit             |
|----------------------------------------------------------------|-----------|------------|----------|--------------|------------------|
| Toggle Frequency (Sine Wave Input)                             | $f_t$     | 0.1        | 1.4      | 1.1          | GHz              |
| Supply Current Output Unloaded (Pin 2)                         | $I_{CC}$  | —          | 4.0      | 6.5          | mA               |
| Modulus Control Input High (MC)                                | $V_{IH1}$ | 2.0        | —        | $V_{CC}$     | V                |
| Modulus Control Input Low (MC)                                 | $V_{IL1}$ | —          | —        | 0.8          | V                |
| Divide Ratio Control Input High (SW)                           | $V_{IH2}$ | $V_{CC}$   | $V_{CC}$ | $V_{CC}$     | Vdc              |
| Divide Ratio Control Input Low (SW)                            | $V_{IL2}$ | Open       | Open     | Open         | —                |
| Output Voltage Swing ( $C_L = 12$ pF; $R_L = 2.2$ k $\Omega$ ) | $V_{out}$ | 1.0        | 1.6      |              | V <sub>pp</sub>  |
| Modulus Setup Time MC to Out                                   | $t_{set}$ | —          | 11       | 16           | ns               |
| Input Voltage Sensitivity 250–1100 MHz<br>100–250 MHz          | $V_{in}$  | 100<br>400 | —<br>—   | 1500<br>1500 | mV <sub>pp</sub> |
| Output Current ( $C_L = 12$ pF; $R_L = 2.2$ k $\Omega$ )       | $I_O$     | —          | 1.5      | 4.0          | mA               |

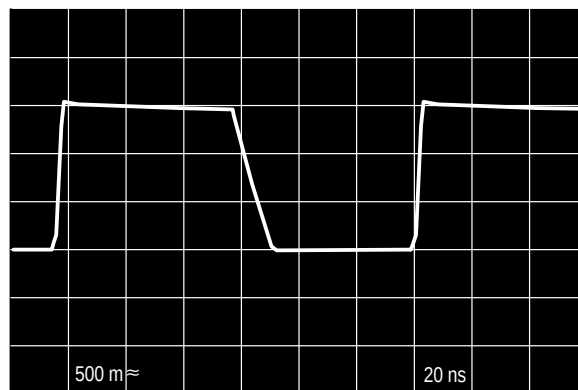
**Figure 1. Logic Diagram (MC12028A)**



**Figure 2. Modulus Setup Time**



**Figure 3. Typical Output Waveform**



# MC12028A MC12028B

Figure 4. AC Test Circuit

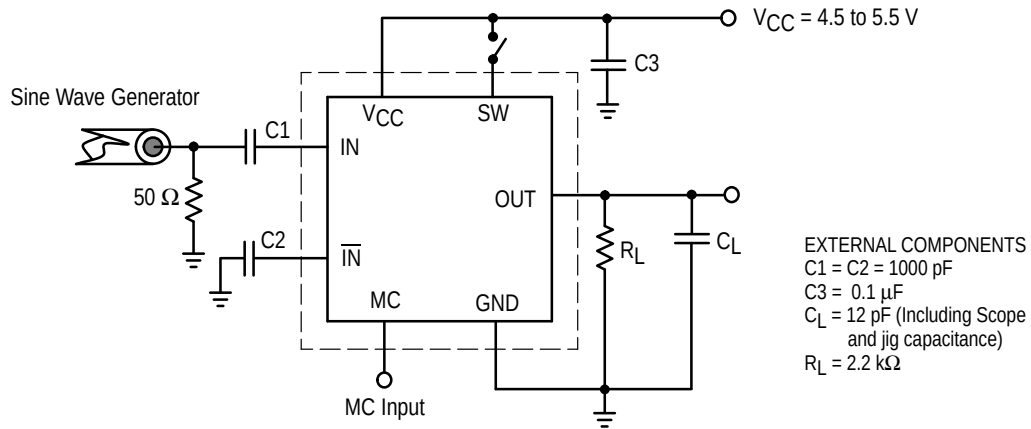
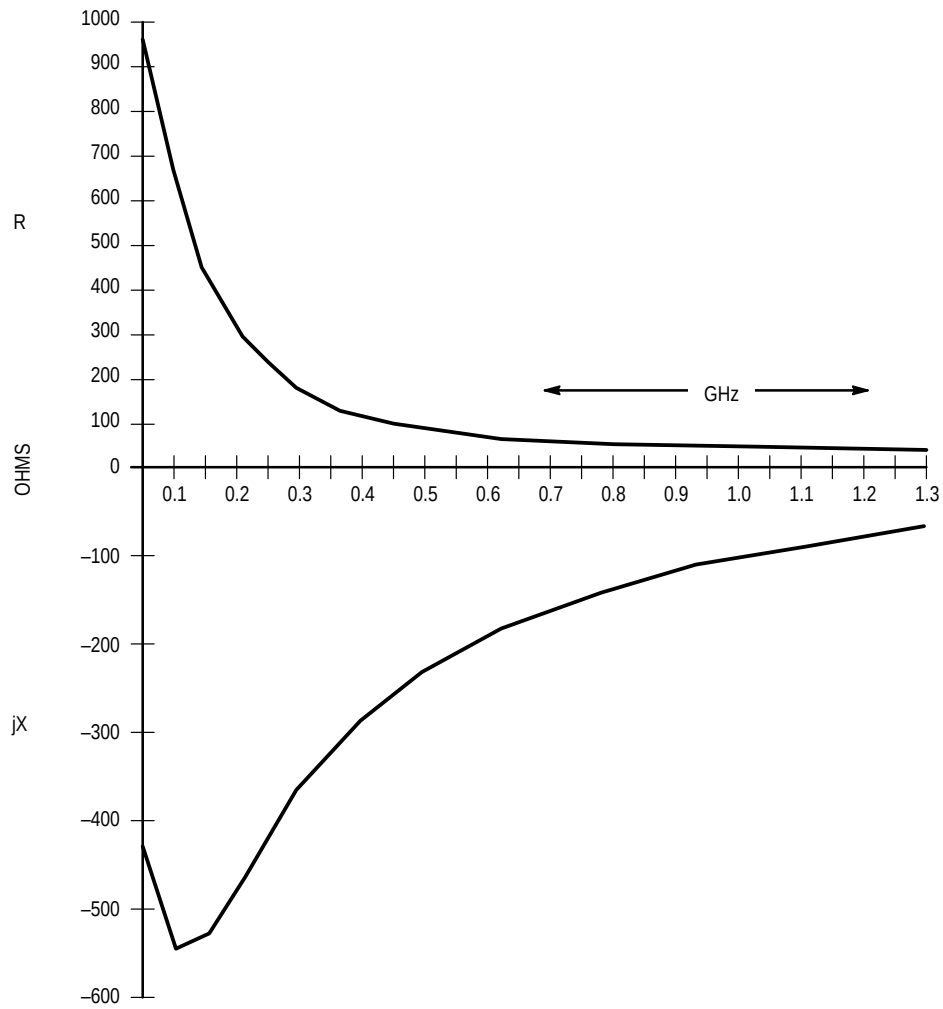
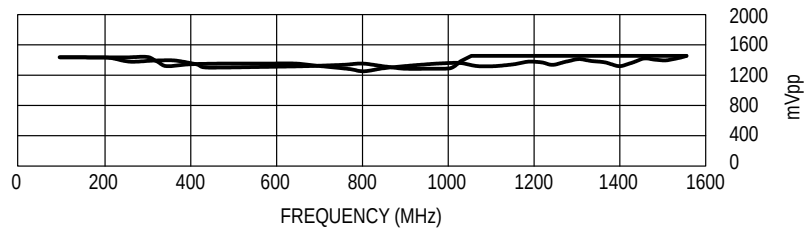
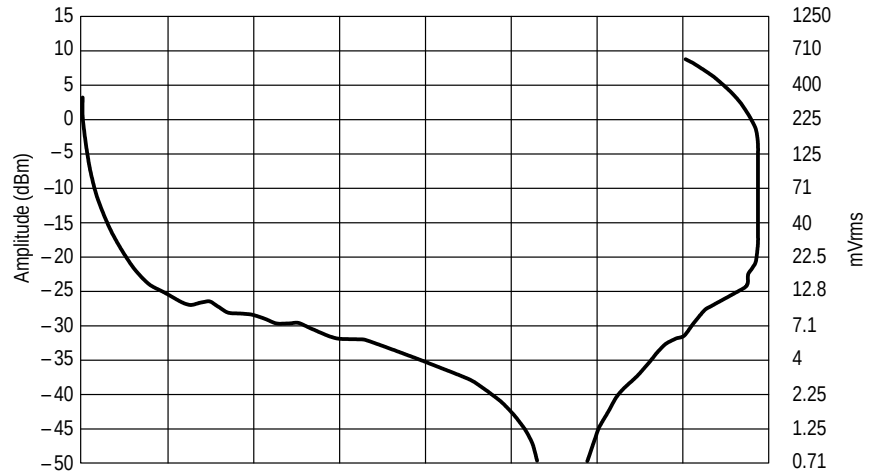


Figure 5. Typical Input Impedance versus Input Frequency



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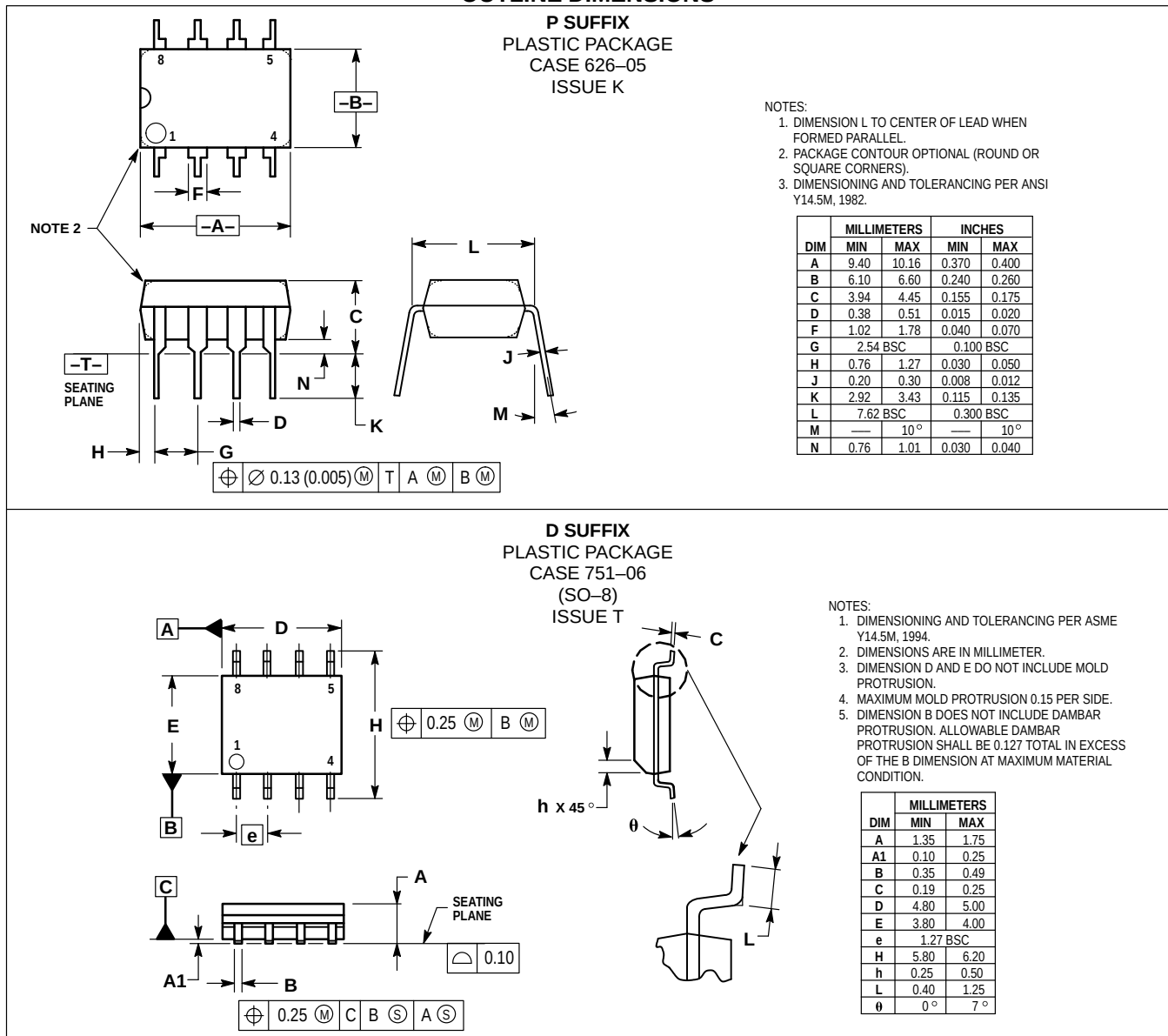
Figure 6. Input Signal Amplitude versus Input Frequency



Divide Ratio = 32

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## OUTLINE DIMENSIONS



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